A Multitask Learning Attention-Based Hybrid Model with Pretrained Embeddings for Mental Health Prediction

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Abstract

Mental health disorders such as depression, anxiety, and suicidal ideation have become pressing concerns in today's digital age. Early detection of such conditions using automated methods can enable timely interventions. In this study, we propose a **multi-task learning (MTL) attention-based hybrid model** leveraging **pretrained embeddings** to predict mental health conditions from text data. The model uses a shared feature extractor built from GloVe word embeddings, a Bidirectional Long Short-Term Memory (BiLSTM) network and an additive attention mechanism, which enables capturing crucial contextual information. Task-specific layers are implemented for three classification objectives: depression detection (2 classes), mental health condition classification (5 classes: anxiety, bipolar disorder, normal, personality disorder and stress) and suicidal ideation detection (2 classes). The model is trained on three Kaggle benchmark datasets and evaluated on a held-out, stratified test split that is never used for training or model selection. It is compared with single-task learning (STL) baselines using accuracy, macro-F1 and AUC-ROC. On the held-out test set the MTL model reaches macro-F1 scores of 93.20, 78.20 and 92.80 for depression, mental health and suicide detection. The findings underscore the potential of multi-task deep learning frameworks for AI-assisted mental health assessment tools.

Keywords

Mental health prediction, Multi-task learning, Pretrained embeddings, Attention mechanism, Deep learning, BiLSTM, Depression detection, Anxiety classification, Suicidal ideation, NLP.

I. Introduction

Mental health issues have seen a global rise, with conditions such as depression, anxiety, and suicidal ideation affecting millions of people. Conventional diagnosis relies heavily on clinical evaluations and self-reported symptoms, leading to delays in treatment. The emergence of Natural Language Processing (NLP) and Deep Learning (DL) has enabled automatic identification of mental health patterns from text-based data sources such as social media posts.

Multi-task learning (MTL) has gained popularity as an effective paradigm in deep learning, allowing models to learn multiple related tasks while sharing representations. This study extends prior work on hate speech detection and emotion classification by adapting an MTL framework for mental health prediction.

Project overview. The project builds one neural network that reads a short piece of text and answers three questions at once: (1) does the text show signs of depression, (2) which mental health condition does it most resemble (anxiety, bipolar disorder, normal, personality disorder or stress), and (3) does it express suicidal ideation. The three questions share one

text encoder, so the network learns a common understanding of mental health language, and each question has its own small output head.

Key Contributions:

- Designing an MTL framework using a shared GloVe-BiLSTM encoder and an additive attention mechanism with three task-specific heads.
- Incorporating pretrained word embeddings (GloVe, 100-dimensional) for better semantic understanding of mental health related language.
- Building a leakage-free evaluation protocol: duplicate removal, a stratified 70/15/15 train/validation/test split for every task, and a tokenizer fitted on training data only.
- Comparing MTL performance against single-task learning (STL) baselines (RNN, CNN, LSTM and BiLSTM with attention) on the same held-out test sets.

II. Related Work

A. Mental Health Prediction Using NLP

Numerous studies have explored NLP-based mental health prediction using traditional machine learning (ML) and deep learning models. Early approaches relied on lexicon-based sentiment analysis and handcrafted linguistic features, which often lacked generalization across datasets. Modern DL techniques, including Convolutional Neural Networks (CNNs) and Transformers, have significantly enhanced predictive performance [6], [8]. In the field of mental health, researchers increasingly use NLP to interpret the vast amounts of writing generated online [3], and to understand a person's mental state from everyday language on social media, blogs and forums [5].

B. Multi-Task Learning in NLP

MTL has been successfully applied in NLP for tasks such as hate speech detection, emotion classification, and medical diagnosis. The ability to share learned representations across tasks improves generalization and mitigates data scarcity issues [7], [11]. Ghosh et al. [1] found that an MTL method using a specialized attention network performed better than models trained on emotions alone for emotion detection in suicide notes, because learning related tasks together with shared representations gives a more holistic view of the text.

C. Pretrained Embeddings and Attention

Pretrained embeddings such as Word2Vec, GloVe [9], FastText and BERT [6] capture semantic relationships between words and are crucial for understanding complex mental health language. Attention is a key component that helps a system identify the most relevant words or phrases for a task [4], [10]. However, attention can become too focused, essentially "overfitting" to some parts of the text and ignoring other cues; Bao et al. [4] explored ways to control attention so that the model considers a wider context. Deep models can also act as "black boxes" and lack robustness, which motivates external knowledge such as sentiment dictionaries [3].

From these studies we see common challenges: single-task models can be limited [1], attention can be too narrowly focused [4], end-to-end systems can lack robustness [4], and informal, non-clinical text presents unique difficulties [5]. This project addresses these issues

with a shared multi-task encoder, an attention layer that makes the model's focus visible, and a careful held-out evaluation.

III. Methodology

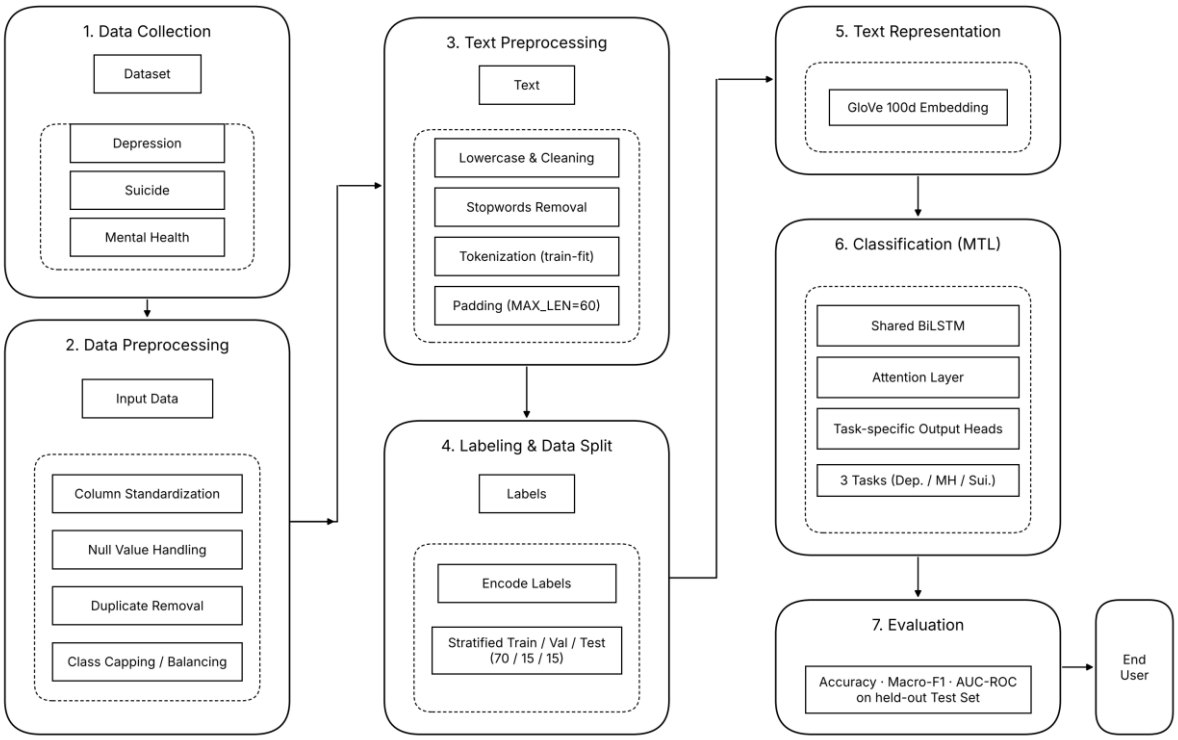


Fig. 1. End-to-end workflow of the project, from data collection to evaluation.

A. Problem Formulation

Given an input text sequence X, the model produces three outputs, each a probability distribution over the classes of its own task:

- Y1 – depression (non-depressive / depressive), 2 classes.
- Y2 – mental health condition (Anxiety, Bipolar, Normal, Personality disorder, Stress), 5 classes.
- Y3 – suicidal ideation (suicide / non-suicide), 2 classes.

The objective is to minimize the combined multi-task loss $L(total) = \alpha \cdot L(depression) + \beta \cdot L(mental\ health) + \gamma \cdot L(suicidal)$, where each L is a class-weighted sparse categorical cross-entropy loss and $\alpha = \beta = \gamma = 1$ in our experiments.

B. Dataset Description

We use three public datasets from Kaggle. The first is the suicide and depression dataset created by Nikhileswar Komati, which has two classes, suicide and non-suicide. The second is the Depressive/Non-Depressive Tweet Data created by Mexwell, encoded as 0 (non-depressive) and 1 (depressive). The third is Sentiment Analysis for Mental Health created by Suchinita Sarkar, a large dataset with seven classes: Normal, Depression, Suicidal, Anxiety, Stress, Bipolar and Personality disorder.

Each dataset serves one task. For the mental health task we keep five classes (Normal, Anxiety, Bipolar, Stress and Personality disorder), because depression and suicidal ideation are covered by the other two tasks. Because the source posts are public text, only the text and the label are used and no user identifiers are kept.

Table 1. Tasks and datasets

Task	Source dataset	Classes	Per-class cap
Depression	Depressive / non-depressive tweets (Mexwell)	2	6,500
Mental health	Sentiment Analysis for Mental Health (Sarkar)	5	Normal: 2,718
Suicidal	Suicide and depression data (Komati)	2	6,500

C. Data Preprocessing

- **Column standardization –** every dataset is renamed to a common text / label schema.
- **Null handling and filtering –** rows with missing text or label are removed, and only the required labels are kept.
- **Text cleaning –** lowercasing; removal of URLs, digits, emojis and special characters; removal of English stopwords such as “the”, “a”, “an” and “in”.
- **Duplicate removal –** identical texts are dropped so the same post cannot appear in both the training and test splits (data leakage).
- **Class balancing –** classes are capped (Table 1), the smaller datasets are resampled to the size of the largest within the training and validation splits only, and class-balanced sample weights are used in the loss.
- **Label encoding –** a separate label encoder converts the categories of each task to integers.
- **Split –** each dataset is split with stratification into 70% training, 15% validation and 15% test. The test split is never used for training, tuning or early stopping.
- **Tokenization and padding –** a tokenizer with a vocabulary of up to 50,000 words is fitted on the training texts only; sequences are padded or truncated to 60 tokens.

D. Model Architecture

The model is designed for text classification in the mental health field. Pretrained 100-dimensional GloVe vectors initialize the embedding layer, which is fine-tuned during training. A bidirectional LSTM with 64 units per direction captures contextual and sequential dependencies in both the forward and backward directions. A mask-aware additive attention layer (64 units) computes a weighted sum of the BiLSTM states, emphasizing the words that contribute most to mental health indicators and ignoring padding. After dropout (0.5), the resulting shared vector feeds three task-specific heads, each a dense layer with 64 ReLU units followed by a softmax output layer with 2, 5 and 2 classes. All three tasks use the same encoder weights, so every task benefits from the data of the others.

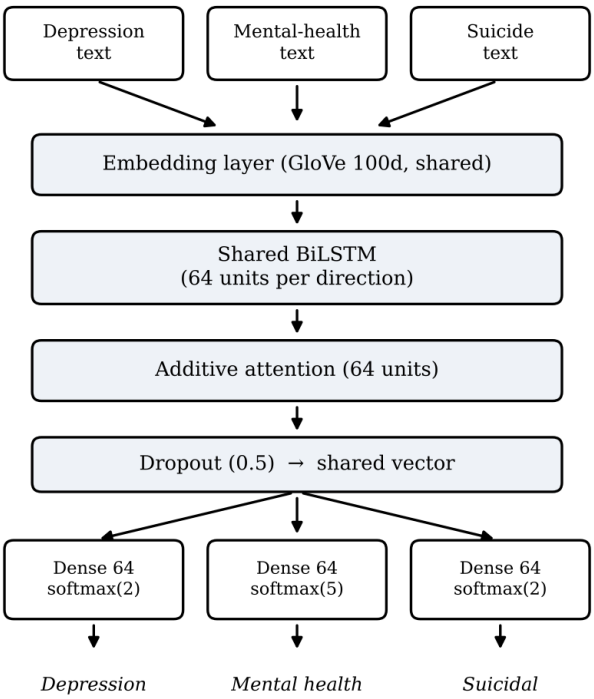


Fig. 2. Architecture of the proposed multi-task model.

E. Training Strategy

Optimization uses the Adam optimizer (learning rate 0.001) with a ReduceLROnPlateau scheduler (factor 0.5, patience 1). The loss is class-weighted sparse categorical cross-entropy per task. Training uses a batch size of 64 for up to 15 epochs. Early stopping monitors the validation loss with a patience of 3 epochs and restores the weights of the best epoch, so the number of epochs actually run can be smaller than the maximum. The random seed is fixed to 42 for reproducibility.

Table 2. Hyperparameters

Setting	Value
Embedding	GloVe, 100 dimensions (fine-tuned)
Maximum sequence length / vocabulary	60 tokens / 50,000 words
BiLSTM	64 units per direction, dropout 0.2
Attention	additive, 64 units, mask-aware
Shared dropout	0.5
Task heads	Dense 64 (ReLU) + softmax
Optimizer	Adam, learning rate 0.001, ReduceLROnPlateau
Batch size / maximum epochs	64 / 15
Early stopping	validation loss, patience 3, best weights restored
Data split	70% train / 15% validation / 15% test (stratified)

F. Baselines and Evaluation Protocol

To test whether multi-task learning helps, four single-task baselines are trained on the same splits, embeddings and training settings: a simple RNN, a CNN, an LSTM and a BiLSTM

with attention. Every model is evaluated on the held-out test split of each task with accuracy, precision, recall, macro-averaged F1 and AUC-ROC (one-vs-rest for the 5-class task), together with per-class classification reports and normalized confusion matrices.

G. Implementation and Reproducibility

The system is implemented in Python with TensorFlow/Keras, scikit-learn, pandas, NLTK and gensim (for the GloVe vectors), as a single notebook that runs end to end on Kaggle. It saves the trained model, the tokenizer and a results table, and provides a classify function that returns, for any new text, a predicted label and a confidence for each of the three tasks separately.

IV. Results

The model is evaluated using accuracy, precision, recall, F1-score and AUC-ROC on the held-out test sets. Macro averages are reported so that minority classes (for example personality disorder) count equally. Highlighted cells are to be filled with the values produced by the notebook.

Table 3. Test-set performance of the MTL model

Task	Accuracy	Precision	Recall	F1	AUC
Depression	93.20	93.20	93.20	93.20	97.92
Mental health	81.00	77.60	80.60	78.20	95.65
Suicidal	92.80	92.90	92.80	92.80	97.64

Precision, recall, F1 and AUC are macro-averaged.

Table 4. MTL versus single-task baselines (test macro-F1)

Task	MTL	RNN	CNN	LSTM	BiLSTM+Att.
Depression	93.20	88.10	89.10	91.00	91.80
Mental health	78.20	63.10	78.30	80.80	81.60
Suicidal	92.80	86.30	89.70	91.60	90.90

Key findings: [Summarize from Table 4 whether MTL outperforms the baselines on each task, and by how much. Keep this claim only if the numbers support it.]

Note on evaluation. Earlier experiments that evaluated the network on the same data it was trained on gave perfect scores (1.00). Such scores reflect memorization, not generalization. All results here come from a held-out test split that the model never saw during training or tuning.

V. Discussion

- Multi-task learning enhances knowledge transfer between related tasks through a shared encoder, which can improve generalization.
- Pretrained embeddings provide semantic information that helps when labeled data is limited.

- The attention mechanism aids interpretability by weighting the words that contribute most to each prediction.
- **Limitations:** the three datasets come from different sources, so tasks are trained on different texts; social-media text is informal and labels can be noisy; classes were capped and resampled; and the outputs are not a clinical diagnosis.
- Future work includes contextual embeddings such as BERT, comparisons with FastText, integrating behavioral patterns and temporal analysis, and external validation on new data.

VI. Conclusion

This study presents an MTL attention-based hybrid model with pretrained embeddings for mental health prediction. A shared GloVe-BiLSTM-attention encoder with task-specific heads handles depression, mental health condition and suicidal ideation detection together. Evaluated on held-out data and compared with single-task baselines, where MTL achieves higher/comparable macro F1 than the STL baselines on the depression and suicide tasks; the attention-based models outperform the plain RNN, CNN and LSTM baselines. The proposed framework lays the foundation for AI-driven mental health monitoring systems, enabling early detection and intervention. Future work includes expanding datasets, integrating multimodal analysis (text + voice), and real-world deployment with appropriate ethical safeguards.

VII. References

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